UNIVERSITY OF ENERGY AND NATURAL RESOURCES, SUNYANI, GHANA

SCHOOL OF SCIENCES

DEPARTMENT OF MATHEMATICS AND STATISTICS

LEVEL 100, END OF FIRST SEMESTER EXAMINATIONS, 2018/2019

BSc. (Comp,Pet,Ren,Elect,Env.Eng,Mech,Agric and Civil) Engineering

MATH 109: ENGINEERING MATHEMATICS 1

December, 2018 **TIME: $2\frac{1}{2}$ HRS**

Index Number: _____

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INSTRUCTIONS

This paper is in two sections A and B

1. **Answer all questions in section A by circling the correct answer on the question paper**
2. **Attempt only one question in section B**

1. Evaluate $\lim_{x\to\infty} \frac{\sqrt{3x^2-16}}{5-2x}$

   A. $\frac{\sqrt{3}}{2}$

   B. $\frac{\sqrt{3}}{5}$

   C. $\frac{-3}{5}$

   D. $\frac{-\sqrt{3}}{2}$

   E. ∞

2. Which of the following is not a solution to the sixth roots of $Z = -8$

   A. $\sqrt{2}\left(\frac{\sqrt{3}}{2} - \frac{1}{2}i\right)$

   B. $\sqrt{2}i$

   C. $\sqrt{2}\left(-\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)$

   D. $\sqrt{2}\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)$

3. Which of the following proposition is not true using the Principle of Mathematical Induction?

   A. $1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2}$

   B. $1 + 3 + 5 + \ldots + (2n - 1) = n^2$

   C. $n^3 - n$ is divisible by 6

   D. $2^n + 6^n$ is divisible by 8

   E. None of the above

4. Which of the following defines a function f for which $f(-x) = -f(x)$?

   A. x^2

   B. $\sin x$

   C. $\cos x$

   D. $\log x$

   E. e^x

5. The complex relation $a + ib > c + id$ can be explained only when
   A. $b = 0$, $c = 0$
   B. $a = 0$, $c = 0$
   C. $b = 0$, $d = 0$
   D. $a = 0$, $d = 0$
   E. None of the above
6. Which of the following statements is true?
   A. A number is rational if and only if its square is rational.
   B. A number is irrational if and only if its square is irrational.
   C. At least one of two numbers x and y is irrational if and only if the product xy is irrational
   D. An integer n is odd if and only if $n^2 + 2n$ is odd.
   E. A number n is odd if and only if $n(n + 1)$ is even
7. The value of $(1 + i)^5(1 - i)^5$ is
   A. -8
   B. 32
   C. 8
   D. 8i
   E. None of the above

Given that $g(y) = \begin{cases} y^2 + 5 & \text{if } y < -2 \\ 1 - 3y & \text{if } y > -2 \end{cases}$
use the above informations to answer the next two questions

8. Evaluate $\lim_{y \to 6} g(y)$
   A. 41
   B. 17
   C. 7
   D. 9
   E. does not exists
9. Estimate $\lim_{y \to -2} g(y)$
   A. 41
   B. 17
   C. 7
   D. 9
   E. does not exists
10. Given that $z_1 = 3 + 2i$ and $z_2 = -1 - 4i$, find $Im(z_1 z_2)$
   A. $5 - 14i$
   B. 14
   C. -14
   D. i
   E. None of the above
11. If $f(x) = x^2 g(x)$, find f' in terms of g'
   A. $2xg(x) + x^2 g'(x)$
   B. $xg(x) + x^2 g'(x)$
   C. $x^g(x) + x^2 g'(x)$
   D. $2xg(x) + 2xg'(x)$
   E. $2xg(x) + xg'(x)$

12. The modulus of two complex numbers z_1 and z_2 are $\sqrt{13}$ and $\sqrt{17}$ respectively. Find $|z_1 z_2|$

    A. $\sqrt{17-13} = 2$

    B. $\sqrt{221}$

    C. $13\sqrt{17}$

    D. $17\sqrt{13}$

    E. None of the above

13. Given $z_1 = 6 - 9i$ and $z_2 = 2i$, find $Re(\frac{z_1}{z_2})$

    A. $-\frac{9}{2} - 3i$

    B. $-\frac{9}{2}$

    C. $3i$

    D. $\frac{9}{2} + 3i$

    E. None of the above

14. Which of the following is **not** true about the absolute value of a real number

    A. $|ab|=|a||b|$

    B. $|a + b + c + ... + m|= |a| + |b| + |c| + ...|m|$

    C. $|a - b| \geqq |a| + |b|$

    D. $|a + b| \leqq |a| + |b|$

    E. None of the above

15. Given the complex numbers z and w, which of the following is not true?

    A. $\overline{zw} = \overline{z}\ \overline{w}$

    B. $\overline{z^n} = \overline{z}^n$, where n is an integer.

    C. $\frac{z}{w} = \frac{zw}{|w|^2}$

    D. $\overline{z + w} = \overline{z} + \overline{w}$

    E. None of the above

16. Given the complex number $z = x + iy$, find e^z

    A. $e^x(\cos y + i \sin y)$

    B. $e^y(\cos x + i \sin x)$

    C. $e^y(\cos x - i \sin x)$

    D. $e^x(\cos y - i \sin y)$

    E. None of the above

Let $f(x) = \begin{cases} \frac{|x-3|}{x-3} & x \neq 3 \\ 0 & x = 3 \end{cases}$

Use the above information to answer the next three questions

17. Find the $\lim_{x \to 3^+} f(x)$

    A. 1

    B. -1

    C. 0

    D. ∞

    E. does not exists

18. Find the $\lim_{x \to 3^-} f(x)$

    A. 1

    B. -1

    C. 0

    D. ∞

    E. does not exists

19. Find the $\lim_{x \to 3} f(x)$

    A. 1

    B. -1

    C. 0

    D. ∞

    E. does not exists

20. The $\lim_{x\to-\infty} \dfrac{e^{4t}-e^{-2t}}{8e^{4t}-e^{2t}+3e^{-t}}$ is

A. $\frac{3}{2}$

B. $\frac{1}{4}$

C. $\frac{3}{4}$

D. $\frac{1}{8}$

Given that $f(x) = xe^{-x^2y^2}$. Use the above information to answer the next four questions

21. Find $f_x(xy)$

A. $e^{-x^2y^2} - 2x^2y^2e^{-x^2y^2}$

B. $e^{x^2y^2} + 2x^2y^2e^{-x^2y^2}$

C. $e^{x^2y^2} - 2x^2y^2e^{-x^2y^2}$

D. $e^{-x^2y^2} + 2x^2y^2e^{-x^2y^2}$

E. $-2x^2y^2e^{-x^2y^2}$

22. $f_{xy}(xy)$ is

A. $6x^2ye^{-x^2y^2} + 4x^4y^3e^{-x^2y^2}$

B. $6x^2ye^{-x^2y^2} - 4x^4y^3e^{-x^2y^2}$

C. $-6x^2ye^{-x^2y^2} - 4x^4y^3e^{-x^2y^2}$

D. $-6x^2ye^{-x^2y^2} + 4x^4y^3e^{-x^2y^2}$

E. None of the above

23. Given $f(x) = x(2x-1)$,. Which of the following value of ε verifies the Rolle's Theorem.

A. o

B. $\frac{1}{2}$

C. $\frac{1}{4}$

D. $\frac{1}{8}$

E. None of the above

24. Given $y = 4\sin 3x \cos 2t$, find $\frac{\partial y}{\partial x}$

A. $12\cos 2t \cos 3x$

B. $-4\cos 2t \cos 3x$

C. $-12\cos 2t \cos 3x$

D. $4\cos 2t \cos 3x$

E. None of the above

25. If $y = 3\tan^4 3x$ then $\frac{dy}{dx}$ is

A. $36\tan^3 3x \sec^2 3x$

B. $12\tan^3 3x \sec 3x$

C. $36\tan^3 3x \sec 3x$

D. $12\tan^3 3x \sec 3x$

E. $12\tan^3 3x \sec^2 3x$

26. Find $\frac{dy}{dx}$, if $y = x^3 e^{-2x+5}$

A. $(3x^2 - x^3)e^{-2x+5}$

B. $(3x^2 - 2x^3)e^{-2x+5}$

C. $(3x^2 + 2x^3)e^{-2x+5}$

D. $(x^2 + x^3)e^{-2x+5}$

E. $(x^2 - x^3)e^{-2x+5}$

27. The differential of the quotient $\frac{2\cos 3x}{x^3}$ is

A. $\frac{6}{x^4}[x\sin 3x + \cos 3x]$

B. $\frac{-6}{x^4}[x\sin 3x - \cos 3x]$

C. $\frac{6}{x^4}[x\sin 3x \cos 3x]$

D. $\frac{-6}{x^4}[x\sin 3x + \cos 3x]$

E. none of the above

28. If a parabola has parametric equations $x = t^2,\ y = 2t$ then $\frac{dy}{dx}$ is

    A. $4t$

    B. t

    C. $\frac{1}{t}$

    D. $\frac{1}{4t}$

    E. none of the above

29. Find the partial derivative of $f(x, y) = x^4 + 6\sqrt{y} - 10$ with respect to x

    A. x^3

    B. $\frac{3}{\sqrt{y}}$

    C. $4x^3 + 6$

    D. $4x^3$

    E. None of the above

30. Change $6e^{2-3i}$ into the form $(a + bi)$

    A. $43.89 + 6.26i$

    B. $43.89 - 6.26i$

    C. $-43.89 + 6.26i$

    D. $-43.89 - 6.26i$

    E. None of the above

31. Given that $2 - 3i = \sqrt{a + bi}$, the value of **a** and **b** is

    A. $\mathbf{a} = 2$ and $\mathbf{b} = -3$

    B. $\mathbf{a} = 4$ and $\mathbf{b} = 9$

    C. $\mathbf{a} = -5$ and $\mathbf{b} = -12$

    D. $\mathbf{a} = 5$ and $\mathbf{b} = -12$

    E. None of the above

32. Evaluate $\frac{2}{(1+i)^4}$

A. $\frac{-1}{2}$

B. $\frac{-i}{2}$

C. $\frac{-i}{4}$

D. $\frac{-1}{4}$

E. $2i$

33. Find $\lim_{x\to 0^+} \frac{\sin x}{\sqrt{x}}$

A. 0

B. 1

C. $\frac{\pi}{2}$

D. -1

E. $\frac{-\pi}{2}$

34. The function $f(x) = e^x$ is

A. periodic

B. even

C. odd

D. neither even nor odd

E. even and odd

35. The function $f(x) = e^x$ can be uniquely decomposed into the sum of an even and an odd functions as

A. $\cosh x$ and $\sinh x$

B. $\cosh x$ and $\sinh x$

C. $\cosh x$ and $\sin x$

D. $\cos x$ and $\sin x$

E. $\cos x$ and $\tanh x$

36. The solution to the inequality $|1-k| > 3$ is

A. [$K > 4$ and $K < 2$]

B. [$K > 4$ and $K < -2$]

C. [$K < 4$ and $K > -2$]

D. [$K < 4$ and $K > 2$]

E. None of the above

37. Which of the following satisfy the inequality $\dfrac{t+1}{3t-6} > 0$

A. $[t > 2$ or $t < 1]$

B. $[t > 2$ or $t < -1]$

C. $[t > 2$ or $t < 1]$

D. $[t > -2$ or $t < -1]$

E. None of the above

38. Which of the following is **not** true about a complex number $z = a + bi$

A. $|Z_1 Z_2 Z_3 ... Z_n| = |Z_1||Z_2||Z_3|...|Z_n|$

B. $|\frac{Z_1}{Z_2}| = \frac{|Z_1|}{|Z_2|}$

C. $|Z| = |\overline{Z}|$

D. $|Z| = (Z\overline{Z})^{\frac{1}{2}}$

E. $|z| = a^2 + b^2$

39. Which of the following is **not** true about the absolute value of a real number

A. $|ab|$=$|a||b|$

B. $|a+b|$= $|a| + |b|$

C. $|a-b| \geqq |a| + |b|$

D. $|a+b| \leqq |a| + |b|$

E. None of the above

40. The following are classified as indeterminate except

A. $\infty - \infty$

B. $0/0$

C. 0^{∞}

D. $1/0$

E. None of the above

Section B

1. (a) The complex number z is defined by $z = a + bi$, where $a, b \in \mathbb{R}$. The complex conjugate is denoted by $\bar{z}$. Show that $|z|^2 = z\bar{z}$ and that $\overline{(z - ki)} = \bar{\bar{z}} + ki$ where $k \in \mathbb{R}$. [4 marks]

   (b) The curve C has the equation $3^{x-1} + xy - y^2 + 5 = 0$. Show that $\frac{dy}{dx}$ at the point $(1, 3)$ on the curve C can be written in the form $\frac{1}{\lambda}\ln(\mu e^3)$, where λ and μ are integers to be found. [4marks]

   (c) Find the number c such that $f(x) = Ax^2 + Bx + C$ satisfy the mean value theorem on the closed interval $[a, b]$. [3marks]

   (d) Find the roots of the equation $z^2 + 4\sqrt{3} = 4i$, giving your answers in the form $re^{-i\theta}$ where $r > 0$ and $0 < \theta 2\pi$. [5marks]

2. (a) The complex number z is defined by $z = \dfrac{9\sqrt{3} + 9i}{\sqrt{3} - i}$. Find showing all your working. An expression for z in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta\pi$. [5marks]

   (b) It is given that $\dfrac{d^n}{dx^n}\left(\dfrac{\ln x}{x}\right) = \dfrac{a_n \ln x + b_n}{x^{n+1}}$ where a_n and b_n depend only on n.

      i. Find a_1, a_2 and a_3. [4marks]

      ii. Use the mathematical induction to establish a formula for a_n. [6marks]

3. (a) Given that $y = \dfrac{x}{1 - x}$, use *Libnitz's rule* for successive differentiation to prove that

$$\frac{d^n y}{dx^n} = \frac{n!}{(1 - x)^{n+1}},$$

   [*Hint*: Use $y = x(1 - x)^{-1}$] [6marks]

   (b) The complex numbers z and w satisfy the equations

$$z + (1 + i)w = i \text{ and } (1 - i)z + iw = 1.$$

   Find z and w in the form $x + iy$, where $x, y \in R$. [5marks]

   (c) Differentiate $y = \tan^{-1}\left(\dfrac{ax - b}{bx + a}\right)$ where $\dfrac{bx}{a} > -1$. [4marks]